

DPC-GNN-Acoustic V4

Physics-as-Forward Graph Neural Network for Real-time CT-to-Ultrasound Synthesis

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Version: V4 — Architecture Redesign

1 Core Innovation

Physics is not a loss function. Physics is the forward pass.

The GNN learns *only* material properties (CT $\rightarrow$ speed of sound / attenuation / scattering).

Wave dynamics are executed deterministically by the Leapfrog integrator. There is no `physics_weight` — this is not a hyperparameter choice, it is an architectural necessity.

❌ V3 (PINN-style, WRONG)

```
Loss = L1 + SSIM + physics_weight  
× wave_residual
```

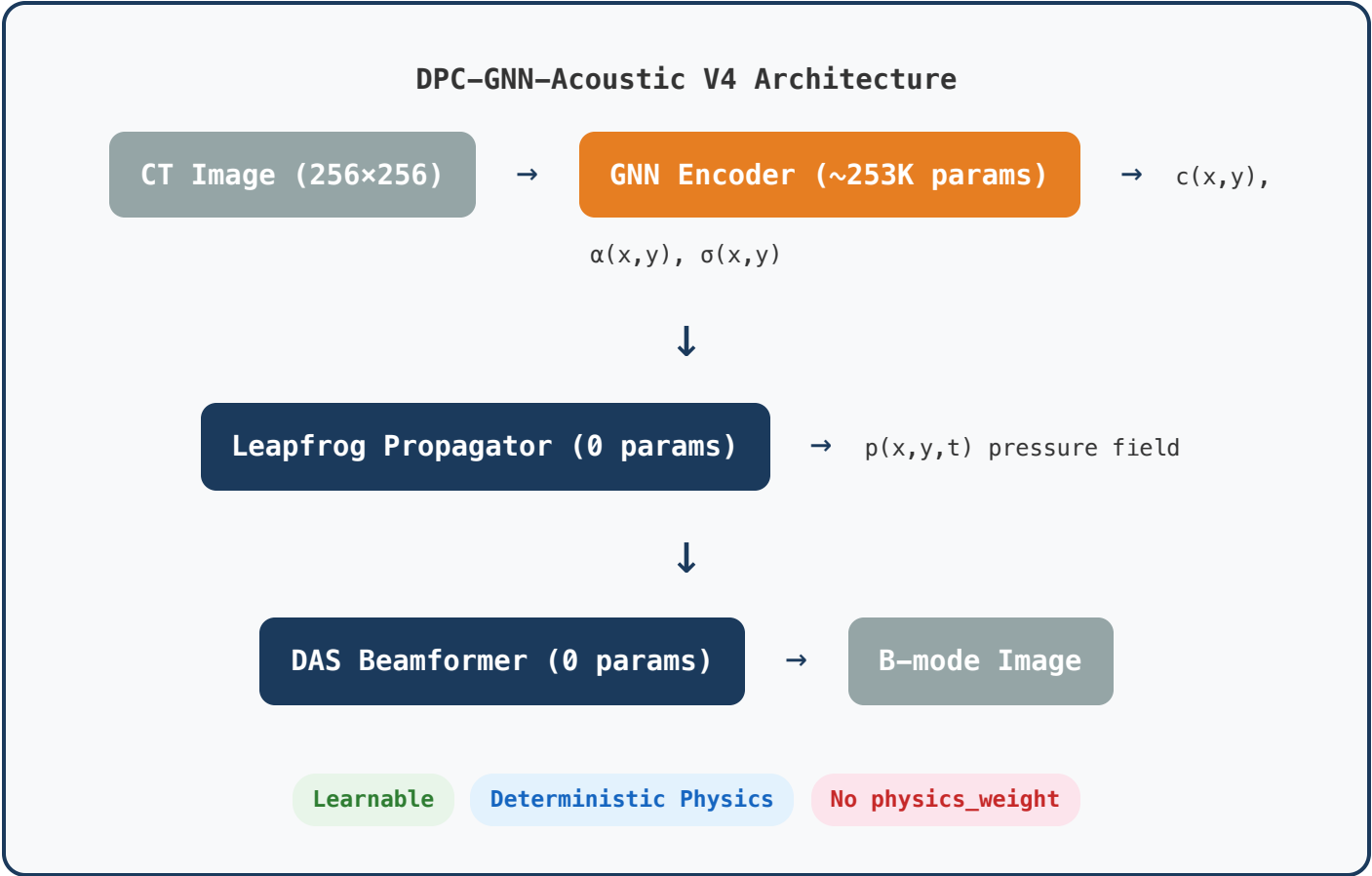
```
# physics_weight = 0.01 ~ 0.1  
# Physics becomes "encouraged" not  
"enforced"  
# GNN directly predicts pressure  
field  
# Leapfrog exists but is bypassed
```

✅ V4 (Physics-as-Forward, CORRECT)

```
Loss = L1 + SSIM    # That's it. No  
physics term.
```

```
# GNN outputs: c(x,y),  $\alpha(x,y)$ ,  
 $\sigma(x,y)$   
# Leapfrog STRICTLY executes wave  
equation  
# Physics residual  $\approx 0$  BY  
CONSTRUCTION  
# physics_weight DOES NOT EXIST
```

2 Architecture



Module Breakdown

Module	Function	Learnable?	Parameters
CNN Downsampler	CT 256 ² → Features 64 ²	Yes	~30K
GNN Processor	5-layer antisymmetric MP, dim=96	Yes	~190K
CNN Upsampler	Features 64 ² → Params 256 ² ×3	Yes	~33K
Leapfrog Propagator	200-step wave equation	No	0
DAS Beamformer	Sensor → B-mode image	No	0
TOTAL			~253K

3 GNN Material Encoder

3.1 Physics-Prior Structure

The speed of sound is decomposed into a known physical prior and a learned residual:

$$c(x,y) = c_{table}(HU(x,y)) + \Delta c_{GNN}(x,y)$$

where c_{table} is a piecewise-linear lookup from established tissue acoustics:

HU Range	Tissue	c_{table} (m/s)	Formula
< -900	Air	343	Constant
-900 ~ -100	Fat/Lung	1430–1540	$1430 + 0.1375 \times (HU + 900)$
-100 ~ 100	Soft tissue	1480–1600	$1540 + 0.6 \times HU$
100 ~ 1000	Bone	1600–3300	$1600 + 1.89 \times (HU - 100)$

GNN learns the residual Δc — spatial context effects, sub-resolution microstructure, and patient-specific variations that lookup tables cannot capture.

3.2 Output Constraints

Parameter	Physical Meaning	Range	Activation
$c(x,y)$	Speed of sound	[1400, 1700] m/s	$\text{sigmoid} \times 300 + 1400$
$\alpha(x,y)$	Attenuation	[0, ~50] Np/m	softplus
$\sigma(x,y)$	Scattering strength	[0, 1]	sigmoid

3.3 Antisymmetric Message Passing

From DPC-GNN soft tissue: $F_{ij} = -F_{ji}$ (Newton's 3rd Law)

In DPC-GNN-Acoustic: $m_{ij} = -m_{ji}$ (Wave equation self-adjointness)

Implementation: $W_{\text{anti}} = W - W^T$, $m_{ij} = \tanh((h_i - h_j) \cdot W_{\text{anti}})$

Property	DPC-GNN (Soft Tissue)	DPC-GNN-Acoustic
Physical law	Newton's 3rd: $F_{ij} = -F_{ji}$	Wave eq. self-adjointness
Predicted quantity	Material properties (μ, λ)	Acoustic parameters (c, α, σ)
Antisymmetry ensures	Momentum conservation	Spatial parameter consistency
Mathematical form	$W_{\text{anti}} = W - W^T$	Same

4 Deterministic Leapfrog Propagator

Wave Equation

$$\partial^2 p / \partial t^2 + 2\alpha(x,y) \cdot \partial p / \partial t = c(x,y)^2 \cdot \nabla^2 p + \sigma(x,y) \cdot s(t)$$

Leapfrog Discretization

$$p^{n+1} = [2p^n - (1 - \alpha\Delta t/2)p^{n-1} + \Delta t^2(c^2\nabla^2 p^n + \sigma s^n)] / (1 + \alpha\Delta t/2)$$

Numerical Stability

Parameter	Value	Status
Grid size	256 × 256	
dx	2.34 × 10 ⁻⁴ m	
dt	2.0 × 10 ⁻⁸ s	
c range	[1400, 1700] m/s	

CFL number	0.206	✅ << 1.0 (very stable)
Points per wavelength	~12 (at 5MHz)	✅ > 6 (low dispersion)
Dispersion error	~0.36%	✅ negligible
Time steps	200	Gradient checkpointed
PML width	20 cells	Cubic damping profile

Key property: The propagator has **zero learnable parameters**. Given (c, α, σ) from the GNN, the pressure field evolution is fully determined by physics. Gradients flow through the propagator via autograd, enabling end-to-end training.

5 Differentiable DAS Beamformer

Standard ultrasound imaging pipeline, fully differentiable:

Step	Operation	Implementation
1. Delay-and-Sum	Time-delay focusing	Linear interpolation (differentiable)
2. Envelope Detection	Hilbert transform	FFT-based (differentiable)
3. Log Compression	Dynamic range	$\log(\text{env} + \epsilon)$, $\epsilon=10^{-6}$ (smooth)

6 Training

Loss Function

```
Loss = L1(pred_bmode, gt_bmode) + (1 - SSIM(pred_bmode, gt_bmode))
```

```
# That's it. No physics term.
```

```
# physics_residual is MONITORED, not TRAINED.
```

Why No Physics Loss?

Proposition 1 (Physics-weight optimality):
For the DPC-GNN-Acoustic architecture, any nonzero physics weight $\lambda > 0$ in the loss $L_{PINN} = L_{data} + \lambda \cdot r^2_{phys}$ is *redundant*, because r_{phys} is identically at numerical precision regardless of θ . The gradient $\nabla_{\theta}(\lambda \cdot r^2_{phys}) \approx \mathbf{0}$ contributes only numerical noise. Therefore, **$\lambda = 0$ is not a hyperparameter choice but an architectural necessity.**

Training Configuration

Hyperparameter	Value
Optimizer	AdamW
Learning rate	1×10^{-4}
Weight decay	1×10^{-5}
Scheduler	Cosine annealing + 5-epoch warmup
Gradient clipping	max_norm = 1.0
Batch size	1 (gradient accumulation: 4)
Epochs	200
AMP	Disabled (numerical stability)

7 DPC-GNN Family Comparison

Aspect	DPC-GNN (Soft Tissue)	DPC-GNN-Acoustic V4
Physical domain	Solid mechanics	Wave propagation
GNN learns	Material properties (μ, λ)	Acoustic params (c, α, σ)
Physics engine	Neo-Hookean + Leapfrog	Wave equation + Leapfrog

Conservation law	Newton's 3rd	Acoustic reciprocity
MP structure	Antisymmetric	Antisymmetric
Physics in loss?	No	No
Learnable dynamics?	No	No
Key result	224× speedup, 0.032mm error	Target: 30×+ speedup, SSIM>0.9

Cross-domain validation: The same architectural principle — "network learns constitutive parameters, physics executes dynamics" — applies across fundamentally different physical domains (solid → wave). This suggests **Physics-as-Forward is a general paradigm**.

8 Experimental Plan

Dataset

Item	Detail
GT source	k-Wave (OMP binary, 256 ²)
Total samples	300 (100 homogeneous + 100 layered + 100 inclusion)
Split	240 train / 30 val / 30 test
Per sample	ct_slice.npy (256 ²) + bmode_gt.npy (128×Nt)
Total size	623 MB

Baselines

Method	Type	Physics	Params
k-Wave	Physical sim	Hard (gold standard)	0
V3 (pw=0.01)	PINN-style	Soft constraint	202K

V3 (pw=0.1)	PINN-style	Stronger soft	202K
U-Net	Pure learning	None	~1M
V4 (Ours)	Physics-as-Forward	Hard (forward)	253K

Target Metrics

Metric	Target	Significance
SSIM	> 0.9	Clinical-quality B-mode
L1	< 0.05	Pixel-level accuracy
Physics residual	≈ 0 ($O(10^{-7})$)	Proves physical self-consistency
Inference time	< 100 ms	Real-time capable
Speedup vs k-Wave	> 30x	Practical clinical value

Ablation Studies

#	Ablation	Purpose
1	Antisymmetric MP vs Standard MP	Spatial consistency contribution
2	c_table + residual vs Direct prediction	Physics prior value
3	Model size: 50K / 150K / 253K / 500K	Capacity sweet spot
4	Time steps: 100 / 200 / 500	Propagation coverage
5	physics_weight: 0 vs 0.01 vs 0.1	Prove $\lambda=0$ is optimal

9 Implementation Status

File	Lines	Status
src/models/dpc_gnn_acoustic_v4.py	344	✅ Complete + Tested

src/models/wave_propagator_v4.py	266	✓ Complete + Tested
src/models/beamform_decoder_v4.py	239	✓ Complete + Tested
src/losses/combined_loss_v4.py	184	✓ Complete + Tested
train_v4.py	368	✓ Complete + Tested
configs/v4_default.yaml	33	✓ Complete
TOTAL	1,434	✓ All verified

Verification Results

Check	Result
Forward pass	✓ B-mode output [1,1,128,128], $c \in [1400, 1700]$
Backward pass	✓ 32/32 encoder params have gradients
Gradient flow	✓ B-mode → beamformer → leapfrog → GNN encoder
Antisymmetric MP	✓ $W - W^T$ enforced
Physics prior	✓ $HU \rightarrow c$ lookup + residual
CFL check	✓ $0.206 < 1.0$

10 Timeline

Phase	Task	Duration
Week 1	V4 smoke test + small-batch validation on GPU	2-3 days
Week 2	Full training (200 epochs) + hyperparameter sweep	5-7 days
Week 3	Ablation experiments + baseline comparisons	5-7 days
Week 4	Paper writing + figures + submission	5-7 days

Generated: March 19, 2026 | "Physics is not a loss. Physics is the forward pass."